



# Modelling Financial Markets

## Seminar 13: Machine Learning in Finance — Hands-on



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## Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AUC: Area Under the (ROC) Curve
- ▣ CI: Confidence Interval
- ▣ CSCV: Combinatorially Symmetric Cross-Validation
- ▣ CV: Cross-Validation
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FFD: Fixed-width window Fractional Differentiation
- ▣ FN: False Negative
- ▣ FRED: Federal Reserve Economic Data
- ▣ GB: Gradient Boosting
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ LW: Local Whittle (estimator)
- ▣ MA: Moving Average
- ▣ MA20: 20-day Moving Average
- ▣ MDA: Mean Decrease Accuracy
- ▣ MDI: Mean Decrease Impurity
- ▣ ML: Machine Learning
- ▣ NN3: Neural Network with 3 hidden layers

## Acronyms used in this material (2/2)

- OLS: Ordinary Least Squares
- OLS-3: OLS on size, book-to-market and momentum only
- PBO: Probability of Backtest Overfitting
- PCR: Principal Component Regression
- PSR: Probabilistic Sharpe Ratio
- PT: Pesaran–Timmermann (test of directional accuracy)
- RC: Reality Check (White's test for data snooping)
- RF: Random Forest
- ROC: Receiver Operating Characteristic (curve)
- RSI: Relative Strength Index
- SD: Standard Deviation
- SE: Standard Error
- SHAP: SHapley Additive exPlanations
- SPA: Superior Predictive Ability (test)
- SR: Sharpe Ratio
- TN: True Negative
- TP: True Positive
- US: United States
- VIX: CBOE Volatility Index

# Outline

Objectives and Setup

Part A: Pen-and-Paper Problems

Part B: Python Problems on Crypto Data

Part C: Open Problem and AI Critique

Assignment and Further Study

## Key Definitions (1/4): Data, Labels, Validation

### Features and labels

- **FFD**:  $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$ , weights truncated when  $|w_k| < \tau$
- **ADF (Dickey & Fuller (1979))**:  $H_0$ : unit root; reject if the statistic is below the 5% critical value (a rejection is not proof of stationarity: A2)
- **EWMA**: standard deviation of daily log returns with exponentially decaying weights (span 50)
- **Triple barrier**: label  $+1/-1$  if the upper/lower barrier ( $pt$ ,  $s/$  multiples of  $\sigma_{t_0} \sqrt{h}$ ) is hit first; else the sign at  $t_0 + h$

### Validation

- **K-fold CV**:  $K$  blocks; fit on  $K - 1$ , test on 1, repeat
- **Leakage**: training data contain information that is also in the test data
- **Purging / embargo**: drop training observations whose label span overlaps the test block, and a few just after it

## Key Definitions (2/4): Models and Classification Metrics

### Models and signals

- ▣ **Logit-L1**: logistic regression with an  $L_1$  (LASSO) penalty
- ▣ **RF**: average of decorrelated trees; **GB**: small trees added sequentially
- ▣ **MA crossover**: long when the fast MA is above the slow one

### Classification metrics

- ▣ Accuracy =  $\frac{TP+TN}{n}$ ; precision =  $\frac{TP}{TP+FP}$ ; recall =  $\frac{TP}{TP+FN}$ ;  $F_1 = \frac{2PR}{P+R}$  (TP, FP, TN, FN: true/false positives/negatives)
- ▣ **ROC AUC**:  $P(\hat{s}_+ > \hat{s}_-) + \frac{1}{2}P(\hat{s}_+ = \hat{s}_-)$  for a random positive and a random negative; 0.5 = no skill
- ▣ **Log loss**:  $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$ , the loss of a probability forecast



## Key Definitions (3/4): Inference and Backtests

### Uncertainty

- ▣ **95% CI**: estimate  $\pm 1.96$  SE
- ▣ **Block bootstrap**: resample blocks of  $L$  consecutive observations to keep serial dependence (Künsch (1989))
- ▣ **HAC**: Newey–West variance (Newey & West (1987)), valid under serial correlation
- ▣ **DM**: Diebold–Mariano test of equal predictive accuracy:  $t$ -test on the loss differential  $d_t = \ell_t^A - \ell_t^B$  with HAC SE (Diebold & Mariano (1995))

### Backtests

- ▣ **SR**:  $\bar{r}/\hat{\sigma}$  of daily log returns; annualised  $\times \sqrt{365}$  for crypto
- ▣ **PSR / DSR** (Bailey & López de Prado (2012); Bailey & López de Prado (2014)): use skewness  $\gamma_3$ , kurtosis  $\gamma_4$  (Normal distribution: 3)
- ▣  $N_{\text{eff}}$ : effective number of independent trials
- ▣ **PBO**: estimated by CSCV (Bailey, Borwein, López de Prado & Zhu (2017))

## How This Seminar Works

- ▣ Two kinds of problems, in every part
  - ▶ **[Solved]**: worked models, with full solutions in this deck and in the notebook
  - ▶ **[Proposed]**: you solve them; each follows an analogous solved problem
- ▣ Part A: pen and paper (derivations, small numerical cases)
- ▣ Part B: Python on Bitcoin, verification *and* inference: every number with its uncertainty
  - ▶ each exercise ends with interpretation questions: what would change the conclusion?
- ▣ Part C: open questions that can seed your team project (C1, C3)
  - ▶ and an AI answer to critique (C2): find the leaks, correct them, log them in `AI_ERRORS.md`
- ▣ Rules: no look-ahead, purged validation, a naive benchmark, and a log of every trial ([Harvey, Liu & Zhu \(2016\)](#))

## Key Definitions (4/4): Memory and Predictive Ability

### Memory

- $I(\delta)$ :  $(1 - B)^\delta X_t$  is  $I(0)$  (stationary, spectral density finite and positive at frequency 0);  $X_t$  itself is stationary only for  $\delta < \frac{1}{2}$
- **LW / ELW**: local Whittle (Robinson (1995)) and exact local Whittle (Shimotsu & Phillips (2005)) estimators of  $\delta$  from the first  $m$  Fourier frequencies;  $SE = 1/(2\sqrt{m})$

### Predictive ability

- **PT**: Pesaran–Timmermann test of sign independence (Pesaran & Timmermann (1992))
- **CW**: Clark–West test for nested models (Clark & West (2007)); **GW**: Giacomini–White test of conditional predictive ability (Giacomini & White (2006))
- **RC / SPA / StepM**: bootstrap tests of the best of many rules (White (2000); Hansen (2005); Romano & Wolf (2005))

## Problem Overview

|   | [Solved]                                                                                                                           | [Proposed]                                                                                                                  |
|---|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| A | A1 FFD weight sums; A3 barrier hitting probabilities; A5 SR variance, PSR, DSR; A7 cost of purging; A9 AUC and its SE              | A2 ADF over-rejection; A4 barriers with drift; A6 DSR and the Gumbel constant; A8 purging; A10 DeLong variance              |
| B | B1 BTC memory; B3 triple barrier; B5 purged CV; B7 models and AUC CI; B9 meta-labelling; B10 DSR and effective $N$ ; B12 PT and CW | B2 ETH memory; B4 asymmetric barriers; B6 find the leak; B8 Diebold–Mariano; B11 PBO; B13 SPA vs. DSR; B14 double selection |
| C |                                                                                                                                    | C1 foundation model vs. boosting; C2 critique an AI answer; C3 ML risk premia on 100 portfolios                             |

- Each proposed problem names its model: the analogous solved problem

## Setup: Python Environment and Data

### Environment

- Google Colab (recommended) or local Python  $\geq 3.10$
- `numpy`, `pandas`, `scipy`, `statsmodels`, `scikit-learn`, `shap`
-  **Open in Colab:**  
`chapter13_seminar_notebook.ipynb`

```
import pandas as pd
url = 'https://raw.githubusercontent.com/danpele/MFM/main/data/market/BTC-USD.CC.csv'
btc = pd.read_csv(url, parse_dates=['date']).set_index('date')['close'].loc['2014-09-17':]
```

### Data

- **Bitcoin in dollars (BTC-USD)**, daily, 2014–2026 (4 385 obs.)
- **Ether (ETH-USD)**, daily, 2015–2026 (4 061 obs.)
- Files in the course repository: `data/market`

## A1 [Solved]: FFD Weights, Their Sums and the Window Length

### Task

- **Context:** fractional differentiation of order  $d$  with weights  $w_0 = 1$ ,  $w_k = -w_{k-1} \frac{d-k+1}{k}$ , truncated at the first  $|w_k| < \tau$  (Hosking (1981); López de Prado (2018)); no data are needed
- **Tasks:**
  - ▶ (a) Show that  $w_k = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)}$  and that  $w_k \sim k^{-(1+d)}/\Gamma(-d)$  for large  $k$ .
  - ▶ (b) Show that  $\sum_{k=0}^K w_k = \frac{\Gamma(K+1-d)}{\Gamma(1-d)\Gamma(K+1)} \sim K^{-d}/\Gamma(1-d)$ .
  - ▶ (c) For  $d = 0.25$  and  $\tau = 10^{-4}$  compute the window  $K$  and  $\sum_{k \leq K} w_k$ .
  - ▶ (d) Decide whether the FFD series of an  $I(1)$  log price is stationary.
- **Report:** the two formulas,  $K$ , the weight sum and the decision in (d)

### Interpretation

- Why do the lecture and the literature call the FFD series “stationary”, and what is actually stationary in it?

## A1: Solution (1/2)

## Solution

- (a)  $w_k$  is the coefficient of  $B^k$  in  $(1 - B)^d = \sum_k \binom{d}{k} (-B)^k$ , and  $(-1)^k \binom{d}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)}$ 
  - Stirling:  $\Gamma(k-d)/\Gamma(k+1) \sim k^{-d-1}$ , so  $w_k \sim k^{-(1+d)}/\Gamma(-d)$ : a hyperbolic, not exponential, decay
- (b) The partial sums of the coefficients of a series  $f(B)$  are the coefficients of  $f(B)/(1 - B)$ ; here  $(1 - B)^d/(1 - B) = (1 - B)^{d-1}$ 
  - so  $\sum_{k \leq K} w_k = \frac{\Gamma(K+1-d)}{\Gamma(1-d)\Gamma(K+1)} \sim K^{-d}/\Gamma(1-d)$ , which tends to 0 only as  $K \rightarrow \infty$

## A1: Solution (2/2)

## Solution

- (c) Set  $|w_K| = \tau$ :  $K \approx (\tau |\Gamma(-d)|)^{-1/(1+d)} = (10^{-4} \cdot 4.902)^{-0.8} = 444$  lags (the recursion also stops at 444)
- Weight sum:  $K^{-d}/\Gamma(1-d) = 444^{-0.25}/1.2254 = 0.2177/1.2254 = 0.178$
- (d) Decompose:  $\tilde{X}_t = (\sum_{k \leq K} w_k) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$ 
  - ▶ if  $X_t$  is  $I(1)$ , every  $X_t - X_{t-k}$  is  $I(0)$ , so the second term is stationary
  - ▶ the first term is  $0.178 X_t$ : a scaled random walk, hence  $\tilde{X}_t$  is still  $I(1)$
- Reading: only the combination  $X_t - X_{t-k}$  is stationary; the “memory” kept by FFD is a small multiple of the price level itself

## A2 [Proposed]: Window Length, Weight Sums and ADF Over-Rejection

### Task

- **Context:** the formulas of A1 with  $\tau = 10^{-4}$ ; a Monte Carlo on pure random walks  $X_t = \sum_{s \leq t} \varepsilon_s$ ,  $\varepsilon_s \sim N(0, 1)$  i.i.d., seed 42
- **Tasks:**
  - ▶ (a) For  $d = 0.3$  and  $d = 0.5$  compute  $K(\tau)$  and  $\sum_{k \leq K} w_k$  with the closed forms of A1.
  - ▶ (b) Simulate 500 random walks, apply FFD with each  $d$  and keep  $T = 2,500$  FFD values per path.
  - ▶ (c) Run the ADF test (constant, 1 lag) on each FFD path and record how often it rejects at 5%.
- **Report:** a table with  $d$ ,  $K$ , the weight sum and the rejection frequency
- **Model:** A1

### Interpretation

- The true size is 5%: why does the ADF confuse a scaled random walk with a stationary series? Which  $d$  would make an  $I(1)$  price stationary?

## A3 [Solved]: Triple-Barrier Labels and Hitting Probabilities

### Task

- **Context:** entry price  $P_{t_0} = 100$ , daily volatility  $\sigma_{t_0} = 1\%$ , vertical barrier  $h = 5$  days,  $pt = sl = 1$ , horizontal barriers  $P_{t_0} e^{\pm \sigma_{t_0} \sqrt{h}}$  (López de Prado (2018)); for (b)–(c) the log price is  $\sigma W_t$ , a driftless Brownian motion
- **Tasks:**
  - ▶ (a) Label path 1 (100, 101.0, 100.4, 102.5, 101.2, 99.8) and path 2 (100, 99.2, 98.6, 99.5, 98.9, 99.1).
  - ▶ (b) With barriers  $+a = pt \sigma \sqrt{h}$  and  $-b = -sl \sigma \sqrt{h}$  and no vertical barrier, derive  $P(\text{upper first})$  and  $\mathbb{E}[\tau]$  by optional stopping.
  - ▶ (c) For  $pt = sl = 1$  compute the probability of reaching the vertical barrier, with continuous and with daily monitoring ( $h = 10$ ).
- **Report:** the two labels, the two formulas and the two time-out probabilities

### Interpretation

- The lecture finds 59.0% time-outs on the S&P 500 and B3 finds 59.0% on Bitcoin: what explains the gap to the Brownian value?

## A3: Solution (1/2)

## Solution

- (a) Width  $0.01\sqrt{5} = 0.02236$ : upper  $100e^{0.02236} = 102.26$ , lower 97.79
  - ▶ path 1: day 3,  $102.5 \geq 102.26 \Rightarrow$  label +1 (the later fall does not matter)
  - ▶ path 2: no horizontal touch;  $\ln(99.1/100) = -0.90\%$  at day 5  $\Rightarrow$  label -1
- (b)  $W_t$  is a martingale:  $0 = p a - (1 - p) b \Rightarrow p = \frac{b}{a+b} = \frac{sl}{pt+sl}$
- $\sigma^2 W_t^2 - \sigma^2 t$  is a martingale:  $\sigma^2 \mathbb{E}[\tau] = p a^2 + (1 - p) b^2 = ab \Rightarrow \mathbb{E}[\tau] = ab/\sigma^2 = pt \cdot sl \cdot h \text{ days (Karatzas \& Shreve (1998))}$

## A3: Solution (2/2)

## Solution

- (c) Exit time of  $|W_s| < 1$  on  $[0, 1]$  (series solution of the heat equation):
  - ▶  $P(\tau > h) = \frac{4}{\pi} \sum_{n \geq 0} \frac{(-1)^n}{2n+1} e^{-(2n+1)^2 \pi^2 / 8} = 1.2732 \cdot 0.2912 - \dots = 0.371$
- Daily monitoring misses touches between closes: shift each barrier out by  $0.5826 \sigma$  (Broadie, Glasserman & Kou (1997)), i.e.  $c = 1 + 0.5826/\sqrt{10} = 1.184$ ; replacing  $\pi^2/8$  by  $\pi^2/(8c^2)$  gives 0.528
- Reading: discrete monitoring explains most of the gap from 37% to 59%; the rest comes from fat tails and volatility clustering (the EWMA volatility lags turbulent periods)

## A4 [Proposed]: Asymmetric Barriers with Drift

### Task

- **Context:** warm-up with  $P_{t_0} = 50$ ,  $\sigma_{t_0} = 2\%$ ,  $h = 4$ ,  $pt = 2$ ,  $sl = 1$ ; then the log price  $X_t = \mu t + \sigma W_t$  with barriers  $+a = pt \sigma \sqrt{h}$  and  $-b = -sl \sigma \sqrt{h}$
- **Tasks:**
  - ▶ (a) Compute both barriers and label the paths (i) 50, 51.5, 53.0, 49.5, 47.9 and (ii) 50, 52.0, 54.5, 53.0, 51.0.
  - ▶ (b) Using the scale function  $s(x) = e^{-2\mu x/\sigma^2}$ , derive  $P(\text{upper first}) = \frac{e^{2\mu b/\sigma^2} - 1}{e^{2\mu b/\sigma^2} - e^{-2\mu a/\sigma^2}}$  and show that it tends to  $\frac{b}{a+b}$  as  $\mu \rightarrow 0$ .
  - ▶ (c) Evaluate the probability for  $\sigma = 3\%$ ,  $\mu = 0.1\%$  per day,  $h = 10$ ,  $pt = 2$ ,  $sl = 1$ .
- **Report:** the barriers, the two labels, the derivation and the number in (c)
- **Model:** A3

### Interpretation

- Formula (b) has no vertical barrier: which paths does a vertical barrier at  $h$  censor, and what benchmark would you compare with B4 for  $(pt, sl) = (2, 1)$ ?

## A5 [Solved]: The Variance of the Sharpe Ratio, PSR and DSR

### Task

- **Context:** i.i.d. returns with mean  $\mu$ , variance  $\sigma^2$ , skewness  $\gamma_3$  and kurtosis  $\gamma_4$ ; a strategy with  $\widehat{SR}_{ann} = 1.2$  on  $T = 756$  daily returns,  $\hat{\gamma}_3 = -0.4$ ,  $\hat{\gamma}_4 = 5$  (Bailey & López de Prado (2012); Bailey & López de Prado (2014))
- **Tasks:**
  - ▶ (a) Derive  $\text{Var}(\widehat{SR}) \approx \frac{1}{T} \left( 1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2 \right)$  by the delta method (Lo (2002)).
  - ▶ (b) Compute  $PSR(0) = \Phi \left( \widehat{SR} \sqrt{T-1} / \sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2} \right)$  with the daily SR.
  - ▶ (c) Compute  $SR_0$  and  $DSR = PSR(SR_0)$  if the strategy is the best of  $N = 100$  independent trials,  $V = 1/T$ .
- **Report:** the variance formula,  $PSR(0)$ ,  $SR_0$  (daily and annualised) and the DSR

### Interpretation

- Would you allocate capital to this strategy? What number of trials  $N$  would make you change your mind?

## A5: Solution (1/2)

## Solution

- (a)  $SR = g(m_1, m_2) = m_1 / \sqrt{m_2 - m_1^2}$  with  $m_j = \mathbb{E}[r^j]$ ; gradient  $\nabla g = (\frac{1}{\sigma} + \frac{\mu^2}{\sigma^3}, -\frac{\mu}{2\sigma^3})$
- Moments of  $(r, r^2)$ :  $\text{Var}(r) = \sigma^2$ ,  $\text{Cov}(r, r^2) = \sigma^3\gamma_3 + 2\mu\sigma^2$ ,  $\text{Var}(r^2) = \sigma^4(\gamma_4 - 1) + 4\mu\sigma^3\gamma_3 + 4\mu^2\sigma^2$
- $T \text{Var}(\widehat{SR}) \approx \nabla g^\top \Sigma \nabla g$ ; the terms in  $\mu^2/\sigma^2$  cancel and leave  $1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2$  (Normal distribution:  $1 + \frac{1}{2} SR^2$ )

## A5: Solution (2/2)

## Solution

- (b) Daily  $\widehat{SR} = 1.2/\sqrt{252} = 0.07559$ ; denominator  $\sqrt{1 + 0.4 \cdot 0.07559 + 1 \cdot 0.07559^2} = 1.01782$ 
  - ▶  $z = 0.07559 \cdot \sqrt{755}/1.01782 = 2.041$ ,  $PSR(0) = \Phi(2.041) = 0.979$
- (c)  $\sqrt{V} = 1/\sqrt{756} = 0.03637$ ;  $\Phi^{-1}(0.99) = 2.326$ ,  $\Phi^{-1}(1 - \frac{1}{100e}) = 2.678$ 
  - ▶  $SR_0 = 0.03637 [0.4228 \cdot 2.326 + 0.5772 \cdot 2.678] = 0.0920$  per day (1.46 annualised)
  - ▶  $z = (0.07559 - 0.0920)\sqrt{755}/1.01782 = -0.444$ ,  $DSR = \Phi(-0.444) = 0.329$
- Decision:  $DSR < 0.95$ ; luck alone gives a best SR of 1.46 > 1.2 among 100 trials: do not allocate

## A6 [Proposed]: DSR on a Shorter Sample and the Gumbel Constant

### Task

- **Context:** a strategy with  $\widehat{SR}_{ann} = 1.5$  on 2 years of daily data ( $T = 504$ ),  $\hat{\gamma}_3 = -0.2$ ,  $\hat{\gamma}_4 = 6$ ;  $N = 20$  independent trials;  $Z_1, \dots, Z_N$  i.i.d.  $N(0, 1)$
- **Tasks:**
  - ▶ (a) Compute  $PSR(0)$ ,  $SR_0$  and the DSR with  $V = 1/T$ , step by step as in A5.
  - ▶ (b) Show that  $\mathbb{E}[\max_{n \leq N} Z_n] \approx (1 - \gamma)\Phi^{-1}(1 - \frac{1}{N}) + \gamma\Phi^{-1}(1 - \frac{1}{Ne})$  in the Gumbel limit, with location  $b_N = \Phi^{-1}(1 - \frac{1}{N})$  and scale  $a_N = \Phi^{-1}(1 - \frac{1}{Ne}) - b_N$ .
  - ▶ (c) Check (b) by Monte Carlo for  $N = 100$  (20,000 draws).
- **Report:**  $PSR(0)$ ,  $SR_0$ , DSR, the derivation and the Monte Carlo mean
- **Model:** A5

### Interpretation

- Compared with A5 the Sharpe ratio is higher and there are fewer trials: is the evidence stronger? What drives the difference?

## A7 [Solved]: How Much Does Purging Cost?

### Task

- **Context:**  $n$  observations in  $K$  contiguous folds of length  $L = n/K$ ; the label of observation  $i$  ends at  $t_1(i) = i + h$ ; embargo of  $e$  observations after each test fold (López de Prado (2018))
- **Tasks:**
  - ▶ (a) For an interior test fold, count the training observations removed before and after the fold, and the share of non-test data lost.
  - ▶ (b) Apply the result to  $n = 20$ ,  $K = 5$ ,  $h = 3$ ,  $e = 1$  and the test fold  $\{8, \dots, 11\}$ ; list the purged, embargoed and training observations.
  - ▶ (c) Explain how the choice of  $K$  trades bias against variance of the CV estimate.
- **Report:** the general formula, the three sets in (b) and the trade-off in (c)

### Interpretation

- Why does shuffled K-fold overstate accuracy in this setting (lecture: 69.4% on pure noise)?

## A7: Solution

## Solution

- (a) Before the fold: observations with  $i + h \geq$  first test index overlap it:  $h$  of them
- After the fold: observations starting before the last test label ends ( $i \leq$  last test index  $+h$ ) and at least the  $e$  embargoed ones:  $\max(h, e)$
- Lost share of the  $n - L$  non-test observations:  $\frac{h + \max(h, e)}{n - L} = \frac{K(h + \max(h, e))}{n(K - 1)}$
- (b) Purged before:  $\{5, 6, 7\}$  ( $t_1 \geq 8$ ); removed after:  $\{12, 13, 14\}$  (12 = embargo, 13–14 start before day 14, when the last test label is known)
  - training set  $\{0, \dots, 4\} \cup \{15, \dots, 19\}$ : 6 of 16 non-test observations lost (37.5%), as the formula says:  $\frac{3+3}{16}$
- (c) Larger  $K$ : more training data per fit (less pessimistic bias), but shorter and noisier test folds and more purged boundaries; when  $h$  is large relative to  $L$ , purging dominates

## A8 [Proposed]: Purging with Shorter Labels

### Task

- **Context:** as in A7, with  $n = 20$ ,  $K = 5$  unshuffled folds, label end  $t_1(i) = \min(i + 2, 19)$  and an embargo of 1 observation
- **Tasks:**
  - ▶ (a) For the test fold  $\{12, \dots, 15\}$ , list the purged observations, the embargoed ones and the training set.
  - ▶ (b) Check your count against the formula of A7.
  - ▶ (c) Use the formula to compute the share of non-test data lost for  $h = 20$  and folds of  $L = 50$  days ( $K = 5$ ).
- **Report:** the three sets and the two shares
- **Model:** A7

### Interpretation

- What does purging cost when labels are long and folds are short, and how would you respond?

## A9 [Solved]: AUC Step by Step, with a Standard Error

## Task

- **Context:** classifier scores of 3 positives (0.80, 0.55, 0.40) and 3 negatives (0.60, 0.35, 0.20); Hanley–McNeil standard error (Hanley & McNeil (1982))
- **Tasks:**
  - ▶ (a) Compute the AUC as the share of (positive, negative) pairs ranked correctly,  $U/(n_1 n_0)$ .
  - ▶ (b) Compute  $SE^2 = \frac{A(1-A) + (n_1 - 1)(Q_1 - A^2) + (n_0 - 1)(Q_2 - A^2)}{n_1 n_0}$  with  $Q_1 = \frac{A}{2-A}$ ,  $Q_2 = \frac{2A^2}{1+A}$ .
  - ▶ (c) Test  $H_0: AUC = 0.5$  at 5% with  $z = (A - 0.5)/SE$ .
- **Report:**  $U$ , AUC, SE,  $z$  and the decision

## Interpretation

- Which assumption of the SE formula fails for overlapping 5-day labels, and in which direction does it bias the test?

## A9: Solution

## Solution

- (a) 0.80 beats 3 negatives, 0.55 beats 2, 0.40 beats 2:  $U = 7$  of 9 pairs,  $A = 0.778$
- (b)  $Q_1 = 0.778/1.222 = 0.636$ ,  $Q_2 = 2 \cdot 0.605/1.778 = 0.681$ 
  - ▶  $SE^2 = [0.173 + 2(0.636 - 0.605) + 2(0.681 - 0.605)]/9 = 0.0429$ ,  $SE = 0.207$
- (c)  $z = (0.778 - 0.5)/0.207 = 1.34$ ,  $p = 0.18$ : do not reject; six observations cannot separate skill from luck
- The formula assumes independent observations; overlapping labels share returns, so the effective sample is smaller and the formula understates the SE: the test over-rejects. B7 replaces it by a block bootstrap

## A10 [Proposed]: AUC as a U-Statistic and the DeLong Variance

### Task

- **Context:** scores of 4 positives (0.70, 0.62, 0.48, 0.45) and 4 negatives (0.66, 0.50, 0.40, 0.30); kernel  $\psi(s_i, s_j) = 1\{s_i > s_j\} + \frac{1}{2}1\{s_i = s_j\}$
- **Tasks:**
  - ▶ (a) Compute  $U$ , the AUC and its Hanley–McNeil SE, as in A9.
  - ▶ (b) Show that  $\text{AUC} = \frac{1}{n_1 n_0} \sum_{i,j} \psi(s_i, s_j)$  is a two-sample U-statistic; define the placement values  $V_{10}(i) = \frac{1}{n_0} \sum_j \psi(s_i, s_j)$  and  $V_{01}(j) = \frac{1}{n_1} \sum_i \psi(s_i, s_j)$ .
  - ▶ (c) Derive  $\widehat{\text{Var}}(\text{AUC}) = S_{10}/n_1 + S_{01}/n_0$  (DeLong, DeLong & Clarke-Pearson (1988)), where  $S_{10}$ ,  $S_{01}$  are the sample variances of the placement values, and compute the DeLong SE.
- **Report:**  $U$ , AUC, the two standard errors and the placement values
- **Model:** A9

### Interpretation

- Why do both formulas assume independent observations, and what does that mean for overlapping 5-day labels?

## B1 [Solved]: Memory of the Bitcoin Log Price: ADF Rule vs. Local Whittle

### Task

- **Context:** daily BTC-USD closes 2014-09-17 to 2026-09-18 (4,385 obs., data/market/BTC-USD.CC.csv),  $x_t = \ln P_t$ ; FFD with  $\tau = 10^{-4}$ ;  $m = n^{0.65}$  Fourier frequencies
- **Tasks:**
  - ▶ (a) For  $d \in \{0, 0.05, \dots, 1\}$  compute the FFD series, the ADF statistic (constant, 1 lag, Dickey & Fuller (1979)) and  $\text{Corr}(\tilde{x}_t, x_t)$ ; find the smallest  $d^*$  for which the ADF statistic is below the 5% critical value.
  - ▶ (b) Estimate the memory with 95% CIs: exact local Whittle (Shimotsu & Phillips (2005)) for  $x_t$  and the FFD series at  $d^*$ , local Whittle (Robinson (1995)) for returns and absolute returns.
  - ▶ (c) Test  $H_0 : d = 1$  for the log price with  $z = (\hat{d} - 1) 2\sqrt{m}$ .
- **Report:** the ADF table for  $d \leq 0.25$ ,  $d^*$ , the four estimates with CIs and the test in (c)

### Interpretation

- Is the FFD series at  $d^*$  stationary? Where does the memory of Bitcoin actually sit?

## B1: Solution (1/2)

### Solution

- (a) ADF statistic along  $d$  (5% critical value  $-2.86$ ):

| $d$  | 0     | 0.05  | 0.10  | 0.15  | 0.20  | <b>0.25</b>  |
|------|-------|-------|-------|-------|-------|--------------|
| ADF  | -1.21 | -2.32 | -2.24 | -2.39 | -2.85 | -3.38        |
| Corr | 1.000 | 0.999 | 0.995 | 0.985 | 0.972 | <b>0.955</b> |

- The rule gives  $d^* = 0.25$  with correlation 0.955; the path is not monotone in  $d$  ( $-2.32$ , then  $-2.24$ ), a first sign of fragility
- The FFD weights at  $d = 0.25$  sum to 0.178, so the FFD series still contains  $0.178 x_t$  (A1)

## B1: Solution (2/2)

### Solution

- (b) Memory estimates,  $m = n^{0.65}$ , 95% CI  $\hat{d} \pm 1.96/(2\sqrt{m})$ :
  - ▶ log price (ELW): 1.065 [1.000; 1.129]; FFD series at  $d^* = 0.25$  (ELW): 0.814 [0.747; 0.880]
  - ▶ returns (LW): 0.063 [-0.002; 0.127]; absolute returns (LW): 0.353 [0.289; 0.417]
- (c)  $z = (1.065 - 1)/0.033 = 1.97$ , two-sided  $p \approx 0.05$ : at most a borderline rejection, in the direction of  $d > 1$  (bubble episodes), never towards stationarity
- Reading: over the band the FFD series has  $\hat{d} \approx 1 - 0.25$  (its integration order is still 1, A1) and its CI lies far above  $\frac{1}{2}$ : it is not stationary; the long memory of Bitcoin is in volatility ( $|r_t|$ ), as for the S&P 500 in the lecture

## B2 [Proposed]: Memory of Ether: a Trap

### Task

- ▣ **Context:** daily ETH-USD closes 2015-08-07 to 2026-09-18 (4,061 obs., data/market/ETH-USD.CC.csv); the functions of B1
- ▣ **Tasks:**
  - ▶ (a) Repeat the ADF rule of B1 on  $\ln P_t^{ETH}$  and report  $d^*$  together with the ADF statistics at  $d = 0$  and  $d = 0.05$ .
  - ▶ (b) Redo the  $d = 0$  test with a constant and a trend, and redo the rule on the sample starting on 1 January 2018.
  - ▶ (c) Estimate  $d$  by exact local Whittle (95% CI) on both samples, and by local Whittle for absolute returns.
- ▣ **Report:** the three ADF-rule answers and the three memory estimates with CIs
- ▣ **Model:** B1

### Interpretation

- ▣ Which  $d$  would you use in a model, and why? What does this say about unit-root tests on an asset that grew 1,000-fold?

## B3 [Solved]: Triple-Barrier Labels on Bitcoin

### Task

- **Context:** daily BTC-USD closes 2014–2026; EWMA volatility of daily log returns (span 50); every day is an event; vertical barrier  $h = 10$  days (López de Prado (2018))
- **Tasks:**
  - ▶ (a) Label every day with symmetric barriers  $pt = sl = 1$  and  $pt = sl = 2$  (units of  $\sigma_{i_0} \sqrt{h}$ ).
  - ▶ (b) Compute the shares of upper, lower and vertical exits, the average holding period and the share of +1 labels.
  - ▶ (c) Compare with the 10-day fixed-horizon label.
- **Report:** a table by setting and a histogram of the holding periods

### Interpretation

- Why are about 56% of the labels +1, and what is then the benchmark accuracy of a classifier?

## B3: Solution

## Solution

- (a)–(b) Exit shares and labels:

|               | Upper | Lower | Vertical | Days | +1    |
|---------------|-------|-------|----------|------|-------|
| $pt = sl = 1$ | 22.8% | 18.2% | 59.0%    | 8.25 | 55.7% |
| $pt = sl = 2$ | 7.2%  | 4.9%  | 87.9%    | 9.58 | 55.6% |
| fixed 10 days | –     | –     | 100%     | 10   | 55.5% |

- (c) Wider barriers push most exits to the vertical barrier: the labels converge to fixed-horizon labels
- Reading: BTC rose over the sample, so +1 dominates whatever the barriers; “always up” (about 56%) is the benchmark to beat, and the 59.0% time-outs are above the 52.8% of A3 (daily monitoring): fat tails and volatility clustering add the rest

## B4 [Proposed]: Asymmetric Barriers on Bitcoin

### Task

- ▣ **Context:** the data, volatility and labelling function of B3,  $h = 10$
- ▣ **Tasks:**
  - ▶ (a) Repeat B3 with  $(pt, sl) = (2, 1)$  and  $(1, 2)$ .
  - ▶ (b) Report the exit shares, the share of +1 labels and the average holding period.
  - ▶ (c) Compare the share of upper exits among horizontal exits with the infinite-horizon value  $\frac{sl}{pt+sl}$  of A3 and with a simulated finite-horizon benchmark (Gaussian daily steps, same barriers, vertical barrier at  $h$ ).
- ▣ **Report:** a table with the two settings and the comparison in (c)
- ▣ **Model:** B3

### Interpretation

- ▣ Which setting would a trend follower choose? How does the choice change the class balance a classifier sees?

## B5 [Solved]: Purged K-Fold vs. Shuffled K-Fold

### Task

- **Context:** a simulated leakage dataset (seed 42): random walk of 3,000 days, label = sign of the 20-day forward return, 5 features = 20-day means of independent noise; true accuracy 50%
- **Tasks:**
  - ▶ (a) Write `PurgedKfold(n_splits, t1, pct_embargo)` with a scikit-learn `split(X)` method.
  - ▶ (b) Fit a random forest ([Breiman \(2001\)](#); 200 trees, minimum leaf 5) and compute the out-of-fold accuracy under shuffled 5-fold, unshuffled 5-fold and purged 5-fold with a 1% embargo.
  - ▶ (c) Check your class on the A7 example.
- **Report:** the three mean accuracies with the five fold values

### Interpretation

- Would the gap shrink with non-persistent features? With 1-day labels?

## B5: Solution

### Solution

- (b) Shuffled K-fold: **69.4%**; unshuffled: 50.0%; purged + 1% embargo: **49.7%**; truth: 50%
- (c) On the A7 example the class removes {5, 6, 7, 12, 13, 14} for the test fold {8, . . . , 11}, as derived step by step
- Reading: shuffling places near-identical neighbours in train and test; a flexible model memorises them and “predicts” pure noise 19 points above chance
- Both ingredients are needed: overlapping labels *and* persistent features; remove either one and the leak disappears (Bergmeir, Hyndman & Koo (2018))

## B6 [Proposed]: Find the Leak

### Task

- **Context:** the B7 modelling frame (BTC, 5-day direction, 11 features, purged 5-fold); the notebook adds a 12th feature `ma11`, and the random forest now reaches a spectacular accuracy
- **Tasks:**
  - ▶ (a) Read how `ma11` is built and rank all features by MDI.
  - ▶ (b) Compute the correlation of `ma11` and of a trailing version with the target return.
  - ▶ (c) Rerun the purged CV with the trailing version and with the 11 base features only.
- **Report:** accuracy and AUC in the three cases, the MDI ranking and the two correlations
- **Model:** B5

### Interpretation

- Why can purging not catch this kind of leak? Give a second example of look-ahead in a feature pipeline.

## B7 [Solved]: Model Comparison and a CI for the AUC

### Task

- **Context:** BTC 2014–2026; target  $y_t = 1\{\ln P_{t+5} - \ln P_t > 0\}$ ; 11 features (momentum 1–120 days, volatility 20/60 and their ratio, distances to the 50- and 200-day moving averages, RSI); purged 5-fold, 1% embargo; 4,181 obs., 54.8% “up”
- **Tasks:**
  - ▶ (a) Fit Logit-L1 (Tibshirani (1996)), a random forest and gradient boosting (Friedman (2001)); report fold-mean accuracy and ROC AUC against “always up”.
  - ▶ (b) Compare MDI, MDA and mean |SHAP| (Lundberg & Lee (2017)) by Spearman rank correlation.
  - ▶ (c) For the pooled out-of-fold RF probabilities compute a 95% CI for the AUC with the DeLong SE (DeLong, DeLong & Clarke-Pearson (1988)) and with a circular block bootstrap (Künsch (1989)),  $L \in \{1, 20, 60\}$ , 1,000 replications.
- **Report:** the model table, the rank correlations and all four intervals

### Interpretation

- Why is the bootstrap interval wider? Which block length would you defend?

## B7: Solution (1/2)

### Solution

- (a) Fold means:

|          | Logit-L1 | RF    | GB    |
|----------|----------|-------|-------|
| Accuracy | 54.8%    | 54.2% | 52.2% |
| ROC AUC  | 0.492    | 0.516 | 0.502 |

- No model beats “always up” (54.8%); Logit-L1 predicts “up” on 99.6% of days
- (b) Spearman: MDI-SHAP 0.75, MDI-MDA  $-0.42$ , MDA-SHAP  $-0.10$ : in-sample and out-of-sample importances disagree

## B7: Solution (2/2)

### Solution

- (c) RF pooled AUC = 0.512
  - ▶ DeLong: SE 0.009, CI [0.495; 0.530],  $z = 1.37$ ,  $p = 0.17$
  - ▶ bootstrap  $L = 1$ : SE 0.009, CI [0.494; 0.530] (same as DeLong)
  - ▶  $L = 20$ : SE 0.015, CI [0.484; 0.543], 21% of replications  $\leq 0.5$ ;  $L = 60$ : SE 0.014, CI [0.485; 0.541]
- Reading: DeLong assumes independent observations; overlapping 5-day labels and volatility clusters violate it, and the dependence-robust SE is 1.7 times larger
- Block length: at least the label horizon (5) and long enough for volatility clusters; results are stable from 20 to 60 days, so  $L = 20$  is defensible

## B8 [Proposed]: Diebold–Mariano on Probability Forecasts

### Task

- **Context:** the pooled out-of-fold probabilities of B7 and a constant forecast equal to the training share of “up” days in each fold; log loss  $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$
- **Tasks:**
  - ▶ (a) Compute the mean log loss of the four forecasts.
  - ▶ (b) For the pairs RF vs. Logit-L1, RF vs. constant and GB vs. constant compute  $d_t = \ell_t^A - \ell_t^B$  and the DM statistic  $\bar{d}/SE_{HAC}(\bar{d})$  (Diebold & Mariano (1995)) with Newey–West lags 4 and 20.
  - ▶ (c) Compare with the naive i.i.d.  $t$ -statistic and decide at 5%.
- **Report:** a table of log losses and of DM statistics with  $p$ -values
- **Model:** B7

### Interpretation

- Can a model with AUC above 0.5 lose to a constant in log loss? What would change the conclusion?

## B9 [Solved]: Meta-Labelling

### Task

- **Context:** BTC 2014–2026; primary model: side of the MA20/MA50 crossover; meta-label = 1 if the primary trade is profitable at its triple-barrier exit ( $pt = sl = 1$ ,  $h = 10$ ) (López de Prado (2018))
- **Tasks:**
  - ▶ (a) Build the meta-labels for every day with a primary signal.
  - ▶ (b) Fit a random forest (balanced class weights) on the features plus the side, on the first 60% of the signals; test on the rest after a 10-day gap.
  - ▶ (c) Compare precision, recall, F1 and the average return per trade before (all trades) and after (only  $\hat{p} > 0.5$ ).
- **Report:** a table with the two rows and the number of trades

### Interpretation

- Precision up, F1 down: which metric matters for a trader who pays costs per trade?

## B9: Solution

### Solution

- Test period 2022-02 to 2026-09, 1,660 signals; the meta-model keeps 49.6% of them

|              | Prec. | Recall | F1    | Trades | Ret./trade |
|--------------|-------|--------|-------|--------|------------|
| Primary only | 0.475 | 1.000  | 0.644 | 1,660  | −0.27%     |
| + meta-model | 0.517 | 0.541  | 0.529 | 824    | +0.19%     |

- Precision rises by 4 points, recall halves and F1 falls; the average return per trade turns from −0.27% to +0.19%
- Reading: with costs per trade, precision and the return per trade matter, not F1; the next step is a test of that return (block bootstrap) and a DSR over all meta-model settings tried

## B10 [Solved]: Grid Search, DSR and the Effective Number of Trials

### Task

- **Context:** long/cash MA crossover on the BTC log price, 15 May 2015 – 2026 ( $T = 4,145$  days, the first day on which every rule has a signal); fast window  $\in \{5, 10, \dots, 55\}$ , slow  $\in \{20, 40, \dots, 240\}$ , slow  $>$  fast:  $N = 120$  trials (Bailey & López de Prado (2014))
- **Tasks:**
  - ▶ (a) Pick the best trial by Sharpe ratio and compute its  $\text{PSR}(0)$ .
  - ▶ (b) From the  $120 \times 120$  correlation matrix of trial returns compute the eigenvalues, the participation ratio  $(\sum \lambda_i)^2 / \sum \lambda_i^2$ , the smallest number of leading eigenvalues whose cumulative sum accounts for 95% of the total and  $\hat{N} = \bar{\rho} + (1 - \bar{\rho})N$ .
  - ▶ (c) Compute  $SR_0$  and the DSR for  $N = 120$ ,  $\hat{N}$  and the 95% count, with  $V = 1/T$ .
- **Report:** the best rule, its SR and moments, the effective- $N$  measures and a DSR table

### Interpretation

- Which  $N$  and which  $V[\widehat{SR}]$  would you report to an investment committee?

## B10: Solution

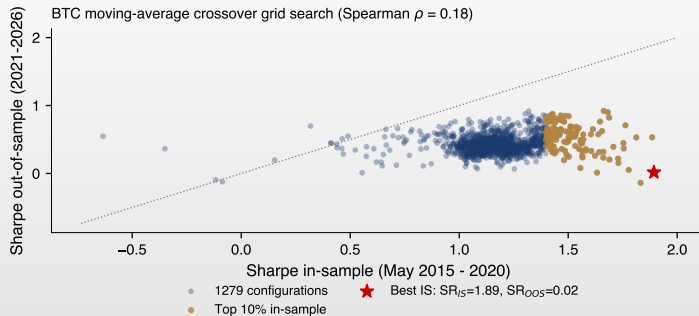
## Solution

- (a) Best (5, 40):  $SR_{ann} = 1.19$  (buy & hold 0.78); skewness 0.13, kurtosis 12.8;  $PSR(0) = 1.000$
- (b)  $\lambda_1 = 105.5$  explains 87.9% of the variance; participation ratio 1.3; 4 eigenvalues reach 95%;  $\bar{\rho} = 0.875 \Rightarrow \hat{N} = 15.9$

| DSR, $V = 1/T$ | $N = 120$ | $\hat{N} = 15.9$ | $N = 4$ |
|----------------|-----------|------------------|---------|
| $SR_0$ (ann.)  | 0.77      | 0.53             | 0.31    |
| DSR            | 0.922     | 0.987            | 0.998   |

- (c) Counting 120 correlated trials as independent over-deflates (0.922); with  $\hat{N}$  the rule passes (0.987)
- Reading: the DSR tests “SR above luck”, not “above buy & hold”; report  $\hat{N}$  with its method, and test against buy & hold separately (B13)

## B10: Why Out-of-Sample Evidence Still Matters



- Lecture result: the best in-sample BTC crossover (SR 1.89, May 2015 – 2020) earned SR 0.02 in 2021–2026

## B11 [Proposed]: Probability of Backtest Overfitting (CSCV)

### Task

- **Context:** the  $T \times 120$  matrix of daily returns of the B10 trials, split into  $S = 16$  contiguous blocks (Bailey, Borwein, López de Prado & Zhu (2017))
- **Tasks:**
  - ▶ (a) For each of the  $\binom{16}{8} = 12,870$  choices of 8 in-sample blocks, pick the trial with the best in-sample Sharpe ratio.
  - ▶ (b) Compute its relative out-of-sample rank  $\omega$  and the logit  $\lambda = \ln \frac{\omega}{1-\omega}$ .
  - ▶ (c) Report  $PBO = P(\lambda \leq 0)$ , the median logit and the median in- and out-of-sample Sharpe ratio of the winner.
- **Report:** PBO, the histogram of  $\lambda$  and the Sharpe ratios in (c)
- **Model:** B10

### Interpretation

- Is the grid overfit? How would  $S$ , or adding 1,000 random strategies to the grid, change the PBO?

## B12 [Solved]: Pesaran–Timmermann and Clark–West on the BTC Direction Models

### Task

- **Context:** the pooled out-of-fold probabilities of B7 ( $n = 4,181$ , 54.8% “up”), predicted sign  $x_t = 1\{\hat{p}_t > 0.5\}$ , and the constant forecast of B8 (Pesaran & Timmermann (1992); Clark & West (2007))
- **Tasks:**
  - ▶ (a) For each model compute  $\hat{P}$ ,  $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$  and the PT statistic.
  - ▶ (b) Because the 5-day labels overlap, compute the slope  $t$ -statistic of  $y_t = a + b x_t + u_t$  with Newey–West errors (4 lags).
  - ▶ (c) Test Logit-L1 against the constant (nested) with the Clark–West statistic on the Brier loss, one-sided, HAC with 4 lags; compare with the plain DM test on the same loss.
- **Report:** a table with  $\hat{P}$ ,  $\hat{P}^*$ , PT and the HAC  $t$ ; the CW and DM statistics with  $p$ -values

### Interpretation

- Why can DM and Clark–West disagree for nested models? Which null does each test?

## B12: Solution (1/2)

## Solution

- (a)–(b) Directional tests:

|          | Pred. up | $\hat{P}$ | $\hat{P}^*$ | PT   | $t_{HAC}$ |
|----------|----------|-----------|-------------|------|-----------|
| Logit-L1 | 99.6%    | 0.548     | 0.548       | 0.89 | 0.82      |
| RF       | 70.3%    | 0.542     | 0.519       | 3.20 | 2.21      |
| GB       | 68.5%    | 0.522     | 0.518       | 0.65 | 0.45      |

- RF: its accuracy (54.2%) is below “always up” (54.8%), yet its signs are not independent of the outcome:  $PT = 3.20$ ; the HAC correction shrinks the evidence to  $t = 2.21$  ( $p = 0.03$ )
- Logit-L1 and GB: no significant directional information

## B12: Solution (2/2)

### Solution

- (c) Clark–West, Logit-L1 vs. constant:  $t = -0.56$ , one-sided  $p = 0.71$ : no evidence that the 11 features add information in the population model
- DM on the Brier loss:  $t = 1.37$  (positive = logit worse),  $p = 0.17$
- DM asks whether the *estimated* logit forecasts better; under the nested null its extra noise makes it lose, so DM rarely favours the larger model
- CW adds back  $(\hat{\rho}_0 - \hat{\rho}_1)^2$  and asks whether the population coefficients are zero; here both agree: no significant signal
- CW is derived for nested linear models and genuine out-of-sample forecasts; on penalised-logit probabilities from pooled purged CV it is an approximate, descriptive adjustment
- Reading: a significant PT (RF) does not make a model useful: its log loss is still worse than the constant (B8)

## B13 [Proposed]: SPA vs. DSR on the 120-Rule Grid

### Task

- **Context:** the  $T \times 120$  return matrix of B10 ( $T = 4,145$ ); benchmarks cash ( $r_0 = 0$ ) and buy-and-hold; stationary bootstrap (Politis & Romano (1994)) with mean block 20 days, 2,000 replications, seed 42
- **Tasks:**
  - ▶ (a) Compute the performance differentials  $d_{k,t} = r_{k,t} - r_{0,t}$  for both benchmarks.
  - ▶ (b) Compute White's Reality Check  $p$ -value (White (2000)), Hansen's SPA  $p$ -value with the consistent recentring (Hansen (2005)) and the number of rules kept by the Romano–Wolf StepM at 5% (Romano & Wolf (2005)).
  - ▶ (c) Compare with the DSR of B10 under  $N = 120$  and  $\hat{N} = 15.9$ .
- **Report:** a table with the largest studentised statistic, the two  $p$ -values and the StepM count per benchmark
- **Model:** B10 and B12; the lecture slide “SPA vs. DSR on the Bitcoin Grid”

### Interpretation

- Which method handles correlated trials without choosing  $N$ ? Does the grid beat buy-and-hold?

## B14 [Proposed]: Double Selection: Does the VIX Predict Bitcoin Returns?

### Task

- **Context:** the B7 frame ( $n = 4,181$ ); outcome  $y_t = 5$ -day forward BTC log return in basis points; regressor  $d_t = \log$  VIX close of the same calendar date, carried forward over weekends and US holidays (data/market/VIX.INDX.csv), standardised; 5 days = 5 BTC calendar days; controls: the 11 features, their squares and pairwise products ( $p = 77$ ), standardised
- **Tasks:**
  - ▶ (a) Run a LASSO of  $y_t$  on  $(d_t, x_t)$  with the plug-in penalty  $\lambda = 2c\sqrt{n}\Phi^{-1}(1 - \gamma/(2p))$ ,  $c = 1.1$ ,  $\gamma = 0.1/\log n$  (Belloni, Chernozhukov & Hansen (2014)); is the VIX kept?
  - ▶ (b) Run the two selection LASSOs ( $y_t$  on  $x_t$ ,  $d_t$  on  $x_t$ ) and report  $|\hat{S}_y|$  and  $|\hat{S}_d|$ .
  - ▶ (c) Estimate OLS of  $y_t$  on  $d_t$  and the union of the selected controls, with HAC errors (4 lags); give  $\hat{\theta}$ , SE and the 95% CI.
- **Report:** the answer to (a), the two set sizes and the post-double-selection estimate
- **Model:** the lecture example “Does the VIX Predict S&P 500 Returns?” and B12

### Interpretation

- Why does the single LASSO give no valid inference? What does the CI say about a crypto “fear premium”?

## C1 [Proposed]: Does a Foundation Model Beat Boosting?

### Task

- **Context:** the BTC 5-day direction of B7 (daily closes 2014–2026); a time-series foundation model of Chapter 14 used zero-shot; baselines of B7/B8; a team project with no single correct method
- **Tasks:**
  - ▶ (a) Choose a model and restrict the test window to dates after its pre-training cut-off, otherwise the test leaks.
  - ▶ (b) Convert the forecast quantiles of the 5-day return into  $P(r_{t,t+5} > 0)$  by interpolation.
  - ▶ (c) Refit gradient boosting and the constant forecast at every forecast origin on a fixed-length rolling window, and compare the log losses on exactly the same dates with the Giacomini–White test ([Giacomini & White \(2006\)](#); HAC errors).
  - ▶ (d) Define a long/cash rule on  $P(\text{up}) > 0.5$  and compute its DSR, or the SPA  $p$ -value, over all configurations you tried.
- **Report:** a Quantlet folder in the team repository (code, `Metainfo.txt`, one chart) and one slide with the conclusion and its main limitation

### Interpretation

- If the foundation model wins, is the gain economically meaningful after costs, and would it survive the next five years of data?

## C2 [Proposed]: Critique an AI Answer

- A student asks an AI assistant: “Predict tomorrow’s direction of Bitcoin from past returns and evaluate the model out of sample with 5-fold cross-validation”
- The assistant answers with this code (btc: daily closes, imports omitted):

```
r = np.log(btc).diff()
X = pd.DataFrame({'r1': r, 'r5': r.rolling(5).sum(), 'vol20': r.rolling(20).std(),
                 'trend10': r.rolling(10, center=True).mean()})
data = pd.concat([X, r.shift(-1).rename('fwd')], axis=1).dropna()
y = (data['fwd'] > 0).astype(int) # 1 if tomorrow is up
Xs = StandardScaler().fit_transform(data[X.columns])
cv = KFold(n_splits=5, shuffle=True, random_state=0)
model = LogisticRegression(penalty='l1', C=0.1, solver='liblinear')
auc = cross_val_score(model, Xs, y, cv=cv, scoring='roc_auc')
print(round(auc.mean(), 3))
```

- “Output: 0.697. An out-of-sample AUC of 0.70 on 4361 days is a strong, tradable signal; cross-validation is out of sample, so there is no overfitting”

## C2 [Proposed]: Critique an AI Answer, Task

### Task

- The answer contains **two** leaks of test information into training, a validation design that does not match deployment and **two** unsupported claims
- **Context:** the code and claim of the previous slide, run on daily BTC-USD closes 2014–2026 (data/market/BTC-USD.CC.csv)
- **Tasks**, for each error
  - ▶ (a) Say what is wrong and why.
  - ▶ (b) Say how you would detect it: a check in the code, a result from B5–B7, a rule of the lecture.
  - ▶ (c) Give the corrected code.
  - ▶ (d) Rerun the corrected code with a walk-forward split and report the AUC.
- **Report:** the five errors as in AI\_ERRORS.md: what was asked, what the AI answered, what was wrong, how you found out, the correct version

### Interpretation

- Which days enter trend10 on day  $t$ ? Which of them is the target?
- Is an AUC of 0.70 for next-day Bitcoin direction plausible, given B6 and B7?
- Why is “cross-validation is out of sample” not enough for time series?

## C3 [Proposed]: Machine Learning Risk Premia on 100 Portfolios

### Task

- **Context:** the lecture case study (Gu, Kelly & Xiu (2020)); the stock panel of the paper is not in the course data, so its design (eqs. 19–21, Section 2.1, Table A.5) is applied to the 100 size–book-to-market portfolios of [Kenneth French Data Library](#)
- **Tasks:**
  - ▶ (a) Build the monthly panel, March 1957 to the latest month: value-weighted excess returns; the return-based characteristics of Table A.6 (mom1m, mom6m, mom12m, mom36m, chmom, maxret, retvol; [Green, Hand & Zhang \(2017\)](#)) and the size and book-to-market deciles, ranked monthly into  $[-1, 1]$ ;  $x_t = (1, \text{tbl}, \text{tms}, \text{dfy}, \text{svar})$  from FRED ([Welch & Goyal \(2008\)](#));  $z_{i,t} = x_t \otimes c_{i,t}$ .
  - ▶ (b) Use the published split (training 1957–1974, validation 1975–1986, test 1987–2016; yearly refits, growing training window, rolling 12-year validation) and fit OLS-3 (the two deciles and mom12m), elastic net, PCR, a random forest and NN3, tuned on validation only.
  - ▶ (c) On 1987–2016 compute the pooled  $R_{OOS}^2$  against a zero forecast (eq. 19) and the DM statistic of each model against OLS-3 (eq. 20, Newey–West errors).
  - ▶ (d) Sort the portfolios monthly into forecast deciles and report the mean, SD and annualised Sharpe ratio of the top-minus-bottom spread; then freeze the design and repeat (c)–(d) from 2017 on.
- **Report:** the results for both periods next to Tables 1, 3 and 7; a Quantlet folder; one slide with the conclusion and main limitation
- **Model:** B7 (models) and B12 (predictive-ability tests)

### Interpretation

- Should  $R_{OOS}^2$  on portfolios exceed that on stocks (Tables 1 and 5), and does the 1987–2016 ranking of models survive after 2016?

## Take-Home Assignment (Team Project)

- Choose one asset (equity index, stock, crypto, exchange rate or commodity) and deliver
  - ▶ features without look-ahead; for any fractional feature, a memory estimate (local Whittle) with its CI, chosen on training data only (B1–B2)
  - ▶ triple-barrier labels with a justified  $(pt, sl, h)$
  - ▶ at least three models with Purged K-Fold, a Pesaran–Timmermann test, a DM or Clark–West test against a constant forecast and a block-bootstrap CI for the AUC
  - ▶ a log of every trial, the effective  $N$ , the DSR, the SPA  $p$ -value against buy-and-hold and the PBO of the final strategy
- GitHub repository with notebooks and Quantlet Metainfo.txt; a 10-minute presentation
- Log every AI error you catch in AI\_ERRORS.md, as in C2: what was asked, what was wrong, how you found out, the correct version
- A negative result, honestly documented, is a good result

## Further Study on Quantinar

-  [Random Forests](#) — bagging, out-of-bag error, variable importance
-  [Shapley Values](#) — model-agnostic local explanations
-  [Machine Learning in Financial Risk](#) — ML estimation of risk measures
-  [Introduction to Reinforcement Learning](#) — Q-learning, sequential decisions

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